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Where predictive information resides in endogenous A-to-I RNA editing

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A-to-I RNA editing helps prevent a cell's own double-stranded RNA from activating antiviral defenses. Yet neighboring adenosines can be edited differently, and editing at the same genomic site can differ across tissues. Existing predictors recognize many of these patterns, but overall accuracy does not separate contributions from the whole RNA duplex, the target neighborhood, and the cellular setting. We asked which gains come from additional biological measurements and which from better use of the RNA information already available. Controlled comparisons in human inverted-Alu duplexes tested both routes against a common sequence–structure reference. Genomic organization added information about the fraction of sites edited within a duplex. Editing differed across tissues at matched sites, but the tested bulk contextual measurements did not consistently improve site-level prediction. Accounting for dependence among sites improved learning, and a correctly centered local representation recovered additional site-level signal from sequence and pairing already present. Richer training targets and more elaborate models mostly yielded limited or adverse returns under the tested regime. Combining whole-duplex and local-augmented predictors, with all choices fixed before testing, improved prediction on unseen duplexes in all five benchmark datasets. The comparisons distinguish improvements obtainable from existing RNA descriptions from tissue-associated variation left unresolved by current measurements. That unresolved variation motivates measuring editing and candidate regulators in the same cells and transcripts.

Significance

Cells edit double-stranded RNA to help prevent their own transcripts from activating antiviral defenses. Editing nevertheless differs among RNAs, neighboring adenosines, and tissues. Predicting these differences is useful, but choosing the next biological experiment requires knowing which information improves those predictions. Genomic organization helped predict how extensively a duplex was edited, and local RNA context helped distinguish its adenosines. Better treatment of correlated sites also improved learning. Available bulk cellular measurements left tissue-associated differences unresolved. Most tested model extensions added little. The results identify gains obtainable from existing RNA descriptions and motivate measuring editing and candidate regulators in the same cells and transcripts.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA). This editing helps prevent a cell's own RNA from triggering antiviral defenses and can alter transcript function (1–5). Most detected human editing sites lie within Alu repeats, abundant repeated sequences whose oppositely oriented copies can fold together within a transcript (6–10). These duplexes provide natural ADAR substrates, but they are not edited uniformly. Neighboring adenosines can have sharply different outcomes, and editing at the same genomic site can differ among tissues (11, 12). Explaining this selectivity requires understanding both the RNA that can be edited and the conditions in which editing occurs.

Predicting likely editing sites is already possible. Sequence-based approaches, including neural networks and fine-tuned language models, recover substantial predictive signal (13–15). AdarEdit extended this approach to a graph of the complete RNA duplex, with nucleotides connected along the sequence and across predicted base pairs (16). These studies also examined motifs and structural sensitivity. Their success leaves unresolved what additional information predicts editing differences between duplexes, among sites within a duplex, and across tissues. An overall prediction score cannot separate these contributions, or tell us which gains come from added biological measurements and which from better use of existing inputs. That distinction matters for deciding what to measure next to understand editing selectivity.

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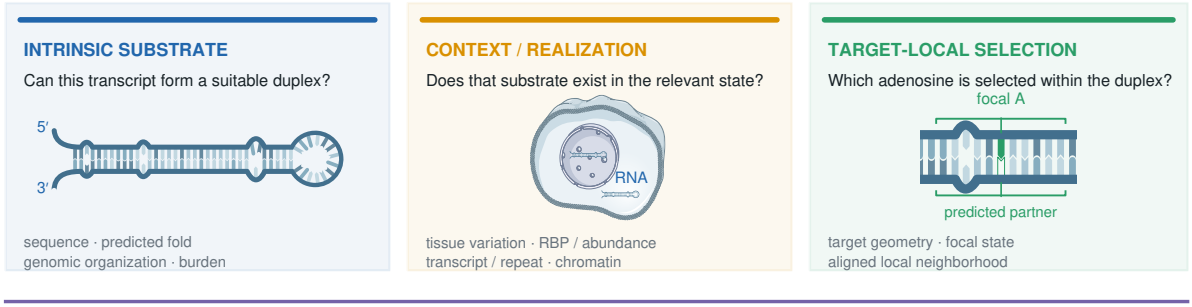
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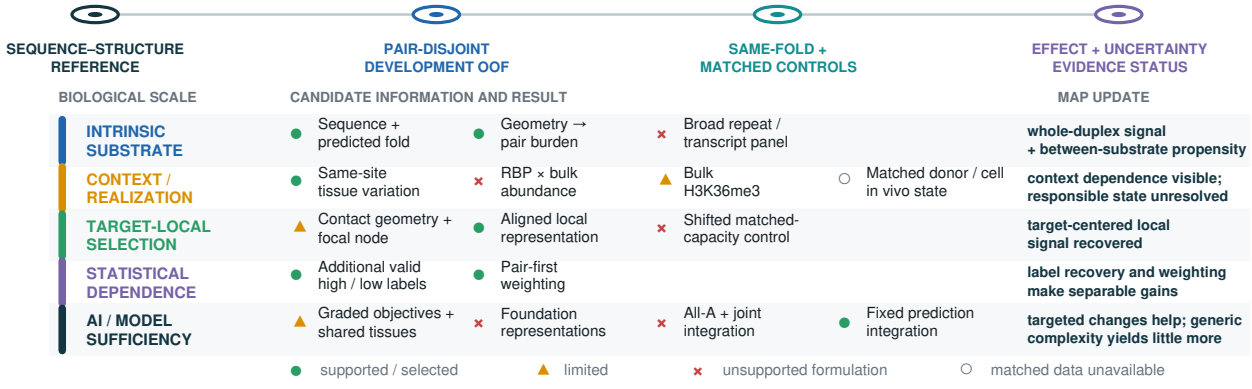
A

The selection problem defines biological and analytical axes



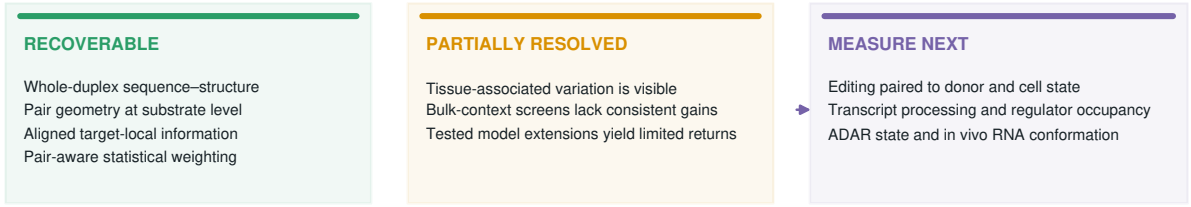
B

Controlled comparisons organize the retained evidence



C

The completed search changes the next question



FROZEN SYNTHESIS CHECK The selected principles improved AUROC, AUPRC, F1, and Brier score in all five held-out endpoints.
unseen benchmark pairs · external-cohort transport remains untested

Fig. 1. Searching for information beyond an already predictive RNA representation. (A) Whole-duplex suitability, transcript/cellular context, and local adenosine selection define the biological questions. Dependence among observations and the sufficiency of model complexity are separate analytical questions. Illustrations are conceptual. (B) Candidates were compared against a sequence–structure reference using pair-grouped development evaluation and relevant matched controls. Symbols distinguish supported/selected, limited, unsupported in the tested formulation, and deferred branches. Early residual screens did not isolate all fitting stages across folds and are exploratory (Methods). (C) Recoverable signal and unresolved context motivate finer matched measurements. The frozen test evaluated only the selected combination on unseen benchmark pairs. SI Appendix, Table S10 records the controls, evidence regimes, and decisions for each branch.

Biochemical and systematic perturbation studies have already identified sequence preferences and structural determinants of editing (17–21). For endogenous RNA, their contributions can be examined at three scales. Whole-duplex sequence, folding, and genomic organization describe a potential substrate (22). Transcript processing, localization, regulator exposure, and enzyme state may determine whether that substrate is available for editing in a cell. Within a duplex, the local sequence and pairing around an adenosine distinguish it from neighboring candidates. Information useful for predicting how extensively a duplex is edited need not identify which of its sites are edited. A static RNA description also need not resolve differences between tissues.

We used the compact AdarEdit sequence–structure model to distinguish additional biological information from better use of existing inputs. Some comparisons added genomic or cellular measurements. Others held the biological inputs fixed while changing how local RNA information was represented or how sites contributed to learning. The weighting comparisons address a separate statistical issue: sites from one duplex share sequence, structure, and context and are not independent observations. Pair-grouped comparisons and relevant matched controls tested incremental predictive value. These increments measure information recoverable under the tested data and representations, not molecular causality.

A parallel search tested whether more detailed training labels, shared learning across tissues, pretrained RNA features, or greater model capacity and integration improved prediction without new measurements (Fig. 1). Results guided which candidates advanced or stopped. We retained limited and unsuccessful outcomes alongside the gains, keeping questions without compatible measurements separate from negative evidence. Only the selected combination was frozen for final held-out evaluation. The Methods and SI Appendix, Table S10 specify the controls and evidence supporting each branch.

Results

Benchmark structure separates discovery from held-out evaluation. We analyzed 884 inverted-Alu pairs with usable sequence–structure graphs and editing measurements from the Genotype-Tissue Expression (GTEx) resource. The five endpoints are artery, brain, liver, skeletal muscle, and the pooled Combined dataset. Each observation is an adenosine within a pair, with a tissue-level editing percentage (EditingIndex). The primary task labels high-editing sites ($\geq$ 15%) as edited and low-editing sites ($<$ 1%) as nonedited; intermediate observations are excluded. A predicted probability therefore refers to the high-editing class, not the percentage of RNA molecules edited.

All sites from each pair stayed together in the published split of 566 training, 139 validation, and 179 test pairs. During development, the training and validation pairs were combined for pair-disjoint out-of-fold (OOF) comparisons: each neural model predicted pairs excluded from its own fitting and selection. Uncertainty was estimated by resampling whole pairs with their sites. Final fitting returned to the published split. Training pairs were used to fit the selected components and validation pairs to select checkpoints and classification thresholds. The frozen specification was then evaluated on the canonical test (Fig. 2). Development comparisons assess individual additions; the held-out test evaluates their selected combination. SI Appendix, Tables S2 and S10 distinguish controlled comparisons, exploratory screens, and questions deferred for unavailable measurements.

Whole-duplex sequence and structure define a strong but non-universal reference. The inherited whole-duplex reference already ranks editing sites well (16). AUROC measures how often edited sites receive higher scores than nonedited sites; it is not a fraction of explained biological variance. Within-context AUROC was 0.869–0.933 across the five human endpoints and 0.823–0.865 across three nonhuman datasets (Fig. 3A). We also assess recovery of edited sites (AUPRC), precision–recall balance after classification (F1), and probability error (Brier score; lower is better). SI Appendix, section A explains these metrics and their units.

Training in one dataset and evaluating in another gave the strongest transfer among human endpoints. Transfer varied more among nonhuman datasets, and human–nonhuman transfer was lower and asymmetric (Fig. 3B and C). Sequence–structure regularities extend across contexts without being universal. These descriptive comparisons span heterogeneous datasets; they neither estimate a common effect nor separate lineage from substrate composition, assay, labeling, and cellular environment. Good site ranking still leaves the contribution of substrate availability, tissue state, and unmeasured conformations unresolved.

Genomic organization adds information about substrate editing burden. Distance between oppositely oriented Alus is already associated with editability (22). The reference RNA input joins the two arms with a fixed separator, omitting their genomic separation. We asked whether this missing information helps predict editing burden beyond graph-derived summaries. Here, burden is the fraction of a pair's eligible high/low sites that meet the high-editing threshold, not the editing percentage at one site.

Greater genomic separation between Alu arms was associated with lower burden in every individual tissue (Spearman $\rho = -0.334$ to -0.482; Fig. 4B). Low-capacity models combining geometry with graph summaries predicted pair-level edited fraction with $R^2 = 0.050$–0.277 and outperformed graph summaries alone in every tissue (Fig. 4C; SI Appendix). Genomic organization thus contributed between-substrate information beyond the tested graph summaries. Duplex formation in cells and ADAR engagement were not measured.

Adding predicted pair burden to a site classifier did not yield a reliable improvement: small ranking gains did not extend consistently to classification or probability accuracy (SI Appendix, section E). Predicting how extensively a duplex is edited does not by itself identify which adenosines are edited.

Editing differs across tissues, but current bulk measurements do not resolve the variation. Editing outcomes varied among tissues even when genomic site identity was fixed. In a post-hoc analysis of development pairs, 1,221 of 74,758 sites observed in at least two individual tissues switched strictly between edited ($\geq$ 15%) and nonedited ($<$ 1%) states.

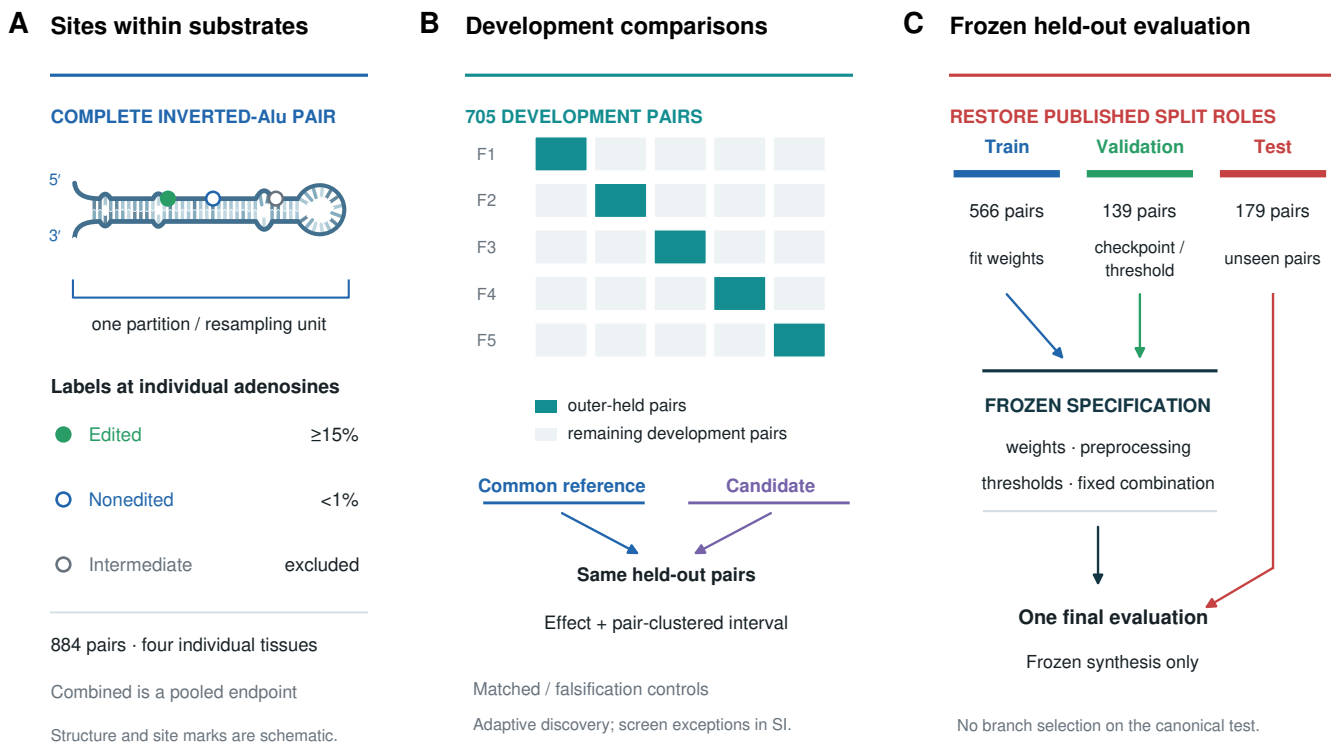


Fig. 2. Complete pairs define the units of development and held-out evaluation. (A) Site labels are nested within 884 inverted-Alu pairs; structure and marked sites are schematic. Intermediate observations are excluded from the canonical task; Combined is pooled. (B) Five pair-disjoint outer folds define development comparisons. Neural candidate and reference models use the same folds, with uncertainty estimated by resampling whole pairs. Matched or falsification controls test specific attributions. Early pooled-score residual screens have the fitting dependency described in Methods. (C) Final fitting restores the published split roles and freezes the specification before evaluating 179 unseen pairs. This test evaluates the selected combination, not each adaptively examined branch.

Among 50,162 sites observed in all four tissues, 17.4% spanned at least 15 EditingIndex points. Rankings of pairs by their observed mean EditingIndex, using identical site sets, still correlated across every tissue pair (ρ = 0.727–0.921; SI Appendix, Fig. S7). Tissue-associated site differences coexist with correlated substrate rankings.

RNA-binding proteins (RBPs) are plausible contributors to these differences because perturbing them can alter editing (23–26). We tested RBP binding maps from cell lines (ENCODE eCLIP), sequence-based binding predictions, and tissue-average RBP expression. Sequence-based predictions complemented eCLIP because repeated Alu sequences make binding reads difficult to assign uniquely. These proxies describe binding potential and average abundance, not exposure of the assayed RNA to a regulator in the same cells.

Combining the RBP proxies yielded no consistent site-level gain beyond the reference and randomized or shuffled controls (27–30). Separate pair-disjoint analyses of matched-site switches and continuous tissue differences found no clear improvement over references using tissue identity alone. The early residual screens reused pooled OOF scores without fully isolating all fitting stages across folds and are exploratory (Methods); the matched-site analyses did not have this dependency.

Transcript and repeat annotations also lacked a stable increment beyond the reference. Bulk overlap with the chromatin mark H3K36me3 yielded metric-specific effects, including a small liver AUPRC increment, without consistent site- or pair-level benefit (31, 32). The tested contextual measurements have not supplied a reliable site-level correction. This negative evidence neither denies the biological role of these regulators nor identifies the source of the remaining error.

Pair-first weighting improves learning from correlated sites. Substrates also differ in how many observed sites they contribute to training. Uniform site sampling gives greater influence to site-rich substrates, although sites in one duplex share sequence, structure, genomic context, and often editing behavior. Balancing that influence tests a statistical improvement without adding a biological measurement. The original class-balanced benchmark also omitted some eligible high/low observations, so label recovery and pair weighting had to be tested separately.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 5A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that pool and class balance fixed, then sampling a pair uniformly within each class before a site, contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). Label recovery and within-class substrate weighting made separable contributions.

The pair-first formulation improved the same-fold reference in all five endpoints (Fig. 5B); all 15 AUROC, AUPRC, and F1 intervals excluded zero. Pair-first sampling is therefore a weighting correction for dependent

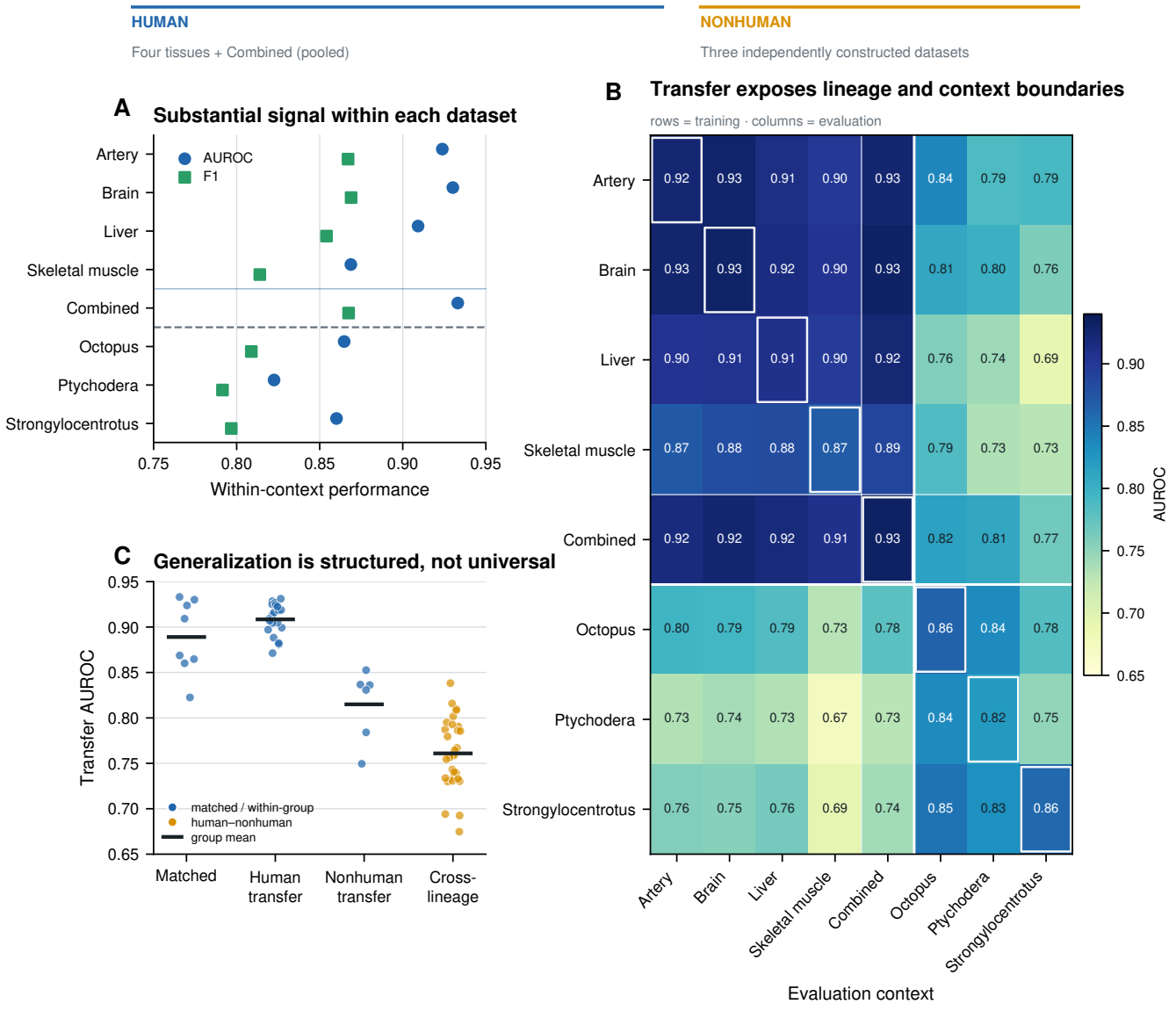


Fig. 3. Sequence and predicted structure carry substantial but non-universal signal. (A) Within-context AUROC and F1 for the retained compact sequence–structure baseline across five human endpoints and three nonhuman datasets: octopus, acorn worm (Ptychodera), and sea urchin (Strongylocentrotus). Combined is pooled. (B) AUROC when a model trained in each context is evaluated in every context; white lines separate human and nonhuman blocks, and outlined diagonal cells mark matched-context evaluations. (C) Transfer grouped by the relation between training and evaluation contexts; horizontal bars are group means. These heterogeneous comparisons are descriptive, not a meta-analysis, and do not identify the cause of transfer differences.

observations in this benchmark, not evidence for an additional biological determinant.

Pair-first learning improved log loss most for pairs with the fewest eligible sites, whereas the two largest site-count quintiles favored row-wise learning (SI Appendix, Fig. S8B).

Target-centered local context adds information beyond whole-duplex pooling. We next asked whether averaging information across a duplex obscures differences between its adenosines. The reference graph already marks the target and contains predicted pairing edges, but pooling averages nucleotide representations into one whole-duplex summary. Exploratory target-contact features improved ranking and reduced Brier error, without a clear F1 gain. Explicitly retaining the target node's representation improved AUROC and AUPRC beyond an otherwise matched control that duplicated the global summary.

The local-augmented predictor adds two aligned 65-position sequence neighborhoods: one around the target adenosine, the other around its predicted pairing partner. A small convolutional encoder reads their nucleotide identities and pairing states and combines the result with the whole-duplex summary. In muscle OOF data, the model reached AUROC/AUPRC/F1 of 0.9045/0.8936/0.8337, compared with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 5C).

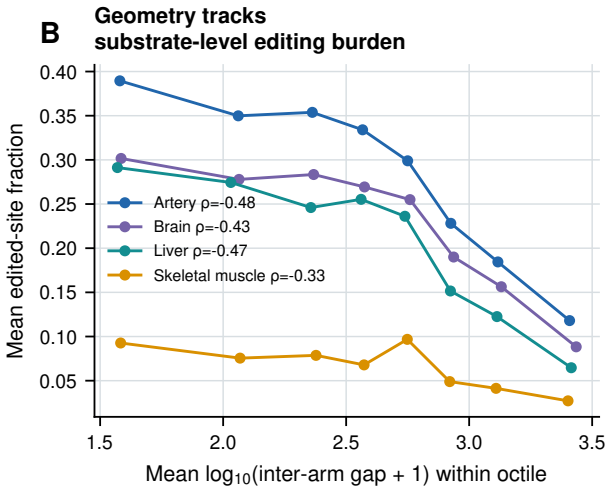
A matched-capacity control retained the same graph-plus-convolution architecture and optimization but shifted the local center by 96 nt. Performance fell to 0.7122/0.6943/0.7028. Aligned-minus-shifted differences were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1,

A

Evidence map: which information changes the recoverable signal?

INFORMATION LEVELS	Tested formulation	Outcome	Interpretation
	LEARNING & SUPERVISION		
LEARNING	Pair-first expanded supervision	● Selected	More labels; correct pair weight
	Continuous + ordinal auxiliary labels	▲ Limited	Small ranking gain
	Shared four-tissue training	▲ Limited	Transfer below preset criteria
	TARGET REPRESENTATION		
TARGET-LOCAL	Target/partner-aligned local view	● Selected	Passes shifted-window control
	Focal node / contact geometry*	▲ Limited	Target-position signal
	Frozen RNA-language embeddings	✖ Not supported	F1-only; no ranking gain
	Substrate-wide all-A decoder	✖ Not supported	Stopped for futility
	Supplied Bio-aware variant	▲ Limited	Features/capacity confounded
	Graph Transformer / GATv2 expansion	✚ Not pursued	Not tested after local gains
	SUBSTRATE & CELLULAR CONTEXT		
SUBSTRATE / CONTEXT	Pair geometry / editing burden	◆ Supported†	Substrate propensity detectable
	Hierarchical engagement gate	▲ Limited	Ranking gain; unstable F1
	Bulk H3K36me3 overlap*	▲ Limited	Metric-specific screen effects
	RBP binding × bulk expression*	✖ Not supported	No consistent screen gain
	Transcript / repeat annotations*	✖ Not supported	No site-level gain
	Matched donor/cell multimodality	■ Deferred	Matched data unavailable
	Matched in vivo structure / isoforms	■ Deferred	Matched data unavailable
INTEGRATION	**INTEGRATION**		
	Fixed whole-duplex + local-augmented logits	● Selected	Complementary predictions
	Unified end-to-end architecture	✖ Not supported	Below frozen ensemble

*Pooled-score exploratory screens. Comparators and intervals: Table S2.



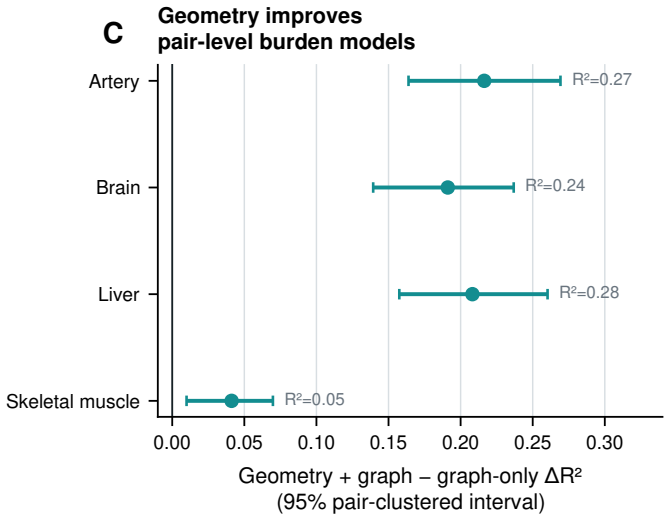


Fig. 4. The evidence map identifies a distinct substrate-level signal. (A) Search outcomes grouped by information scale. SI Appendix, Table S2 gives comparators, effects, and intervals. "Supported†" denotes an observation not yet useful for site prediction; "not supported" concerns the tested formulation. Asterisks mark pooled-score exploratory screens (Methods); within the focal/contact row, only the contact-feature screen has this limitation. Deferred measurements were unavailable. (B) Mean fraction of high-editing sites among eligible high/low sites per pair, across inter-arm-gap octiles; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 after adding geometry to graph summaries. Unlike site-ranking AUROC, R^2 measures prediction of variation in editing burden across pairs. Bars are 95% pair-clustered intervals; labels give combined-model R^2.

with entirely positive pair-clustered intervals. The gain depended on correctly centering the pairing-organized neighborhood, not simply adding a convolutional encoder. Because the full window moved, the control does not isolate the predicted partner from that neighborhood.

The aligned model's advantage over whole-duplex prediction also held in liver and in a secondary evaluation retaining all eligible high/low observations without class downsampling (SI Appendix, section J). The shifted-window control was tested in muscle.

Descriptive Brier improvements occurred for paired and unpaired targets and across every examined contiguous-stem-length stratum (SI Appendix, Fig. S8C and D), so the gains were not confined to one structural stratum.

Whole-duplex and target-local views remain complementary. We combined whole-duplex and local-augmented predictions by averaging their log-odds scores (logits). AUROC/AUPRC/F1 gains over whole-duplex-only prediction were 0.0256/0.0308/0.0139 in muscle and 0.0182/0.0214/0.0179 in liver. All six pair-clustered intervals excluded zero (Fig. 5D).

The final predictor uses two independently trained pair-first components: a whole-duplex-only graph predictor and a graph predictor augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed 0.5/0.5 mean, with no weight search (Fig. 5E). This retains both predictive views without fitting another combination model to only hundreds of independent substrates.

In a post-hoc analysis, edited sites were ranked against nonedited sites within the same duplex, with each mixed-label duplex given equal weight. The fixed ensemble improved mean within-duplex AUROC over the pair-first component in muscle (+1.78 points; 95% pair-bootstrap interval +1.21 to +2.40) and liver (+1.38 points; +1.06 to +1.70). The aligned-local component alone showed a clear within-duplex increment only in muscle, so this analysis supports complementary within-substrate ranking rather than a replicated local-component effect (SI Appendix, Fig. S8A).

Most tested model-expansion routes yield limited returns. The remaining errors could reflect limitations of learning as well as missing measurements. Four classes of model-sufficiency tests examined whether improvements could come from more detailed training labels, shared learning across tissues, richer or pretrained RNA representations, or greater capacity and integration. Each comparison used its corresponding reference. Recovering additional valid high/low labels helped, but training on ordered editing categories and continuous editing levels added only a small ranking gain. Adding that model to the selected combination reduced F1 in the undownsampled high/low pool. Training the four tissues jointly produced limited transfer and worse mean Brier score (Fig. 4A; SI Appendix).

Better representations helped when they preserved target-specific information. Focal-node retention and aligned local encoding improved prediction; fixed features from RiNALMo, a pretrained RNA sequence model, did not add ranking signal beyond matched controls (33). Tighter integration also offered no advantage in the tested comparisons: predicting all adenosines in a duplex jointly and training global and local views in one connected model underperformed their respective references. Burden-based formulations remained limited. SI Appendix separates these completed tests from variants that changed features and architecture together and from untested directions. Within the completed controlled comparisons, substrate-aware weighting and target-centered representation supplied clearer gains than generic model extensions. This result does not define a ceiling for future architectures.

The selected combination improves prediction on unseen benchmark pairs. The final test asked whether the selected learning and representation changes improved prediction together on unseen benchmark pairs. The complete predictor was frozen before canonical held-out evaluation. It improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 6A), with mean gains of 0.0277, 0.0261, and 0.0253 over reproduced supplied-checkpoint AdarEdit. Every endpoint had a positive 95% pair-clustered interval for all three metrics, and Brier score decreased throughout (Fig. 6B and C; SI Appendix, Table S1).

Combined overlaps the individual tissue endpoints, so the mean is descriptive rather than a five-tissue meta-analysis. This test supports generalization of the selected combination to unseen pairs within the benchmark. Attribution to individual components rests on the development comparisons above.

At frozen validation-selected thresholds, the selected predictor corrected 32.0–50.6% of baseline false positives while converting 1.7–5.9% of baseline true positives into false negatives (Fig. 6D). A secondary matched-recall analysis is reported in SI Appendix, Table S9.

Discussion

Genomic organization and target-local RNA context supplied different kinds of predictive information. Earlier studies linked Alu geometry to aggregate editing and identified local sequence and structural preferences through biochemical assays, systematic perturbations, and prediction (17–22). Here, geometry supplied information absent from the fixed-separator RNA input: it improved burden prediction beyond graph summaries but did not yield a reliable site-level correction. Local encoding improved site prediction using sequence and pairing already present in the reference. The shifted-window comparison supports correct centering of that information, without isolating the predicted partner or establishing a molecular mechanism. Pair-first weighting supplied a separate statistical gain by changing the influence of correlated observations.

Context was less resolved. Editing at the same genomic sites differed among tissues even while substrate-level rankings remained correlated. Perturbation studies implicate RBPs and chromatin-associated regulation in editing (23–26, 31), yet cell-line occupancy, tissue-average abundance, transcript and repeat annotations, and bulk chromatin supplied limited predictive value here. These comparisons motivate measuring regulator exposure and transcript state alongside editing at the same biological resolution. They do not identify a latent cellular state or establish that coarse measurement is the sole reason the tested proxies underperformed.

Pair-first learning and target-centered encoding supplied clearer gains than the generic model extensions tested here. Graded supervision and shared-tissue learning gave limited gains; frozen RNA features, substrate-wide decoding, and connected integration did not outperform their respective references on the selection criteria. The fixed combination of the selected components improved held-out discrimination, classification, and probability accuracy. Generic model sophistication alone did not close the gap in the tested regime. Residual error could still reflect finite data, model misspecification, measurement noise, structural uncertainty, unmeasured transcript or cellular state, or biological stochasticity. The present data cannot apportion these sources.

The benchmark contains human inverted-Alu substrates for which graphs could be constructed, with tissue-

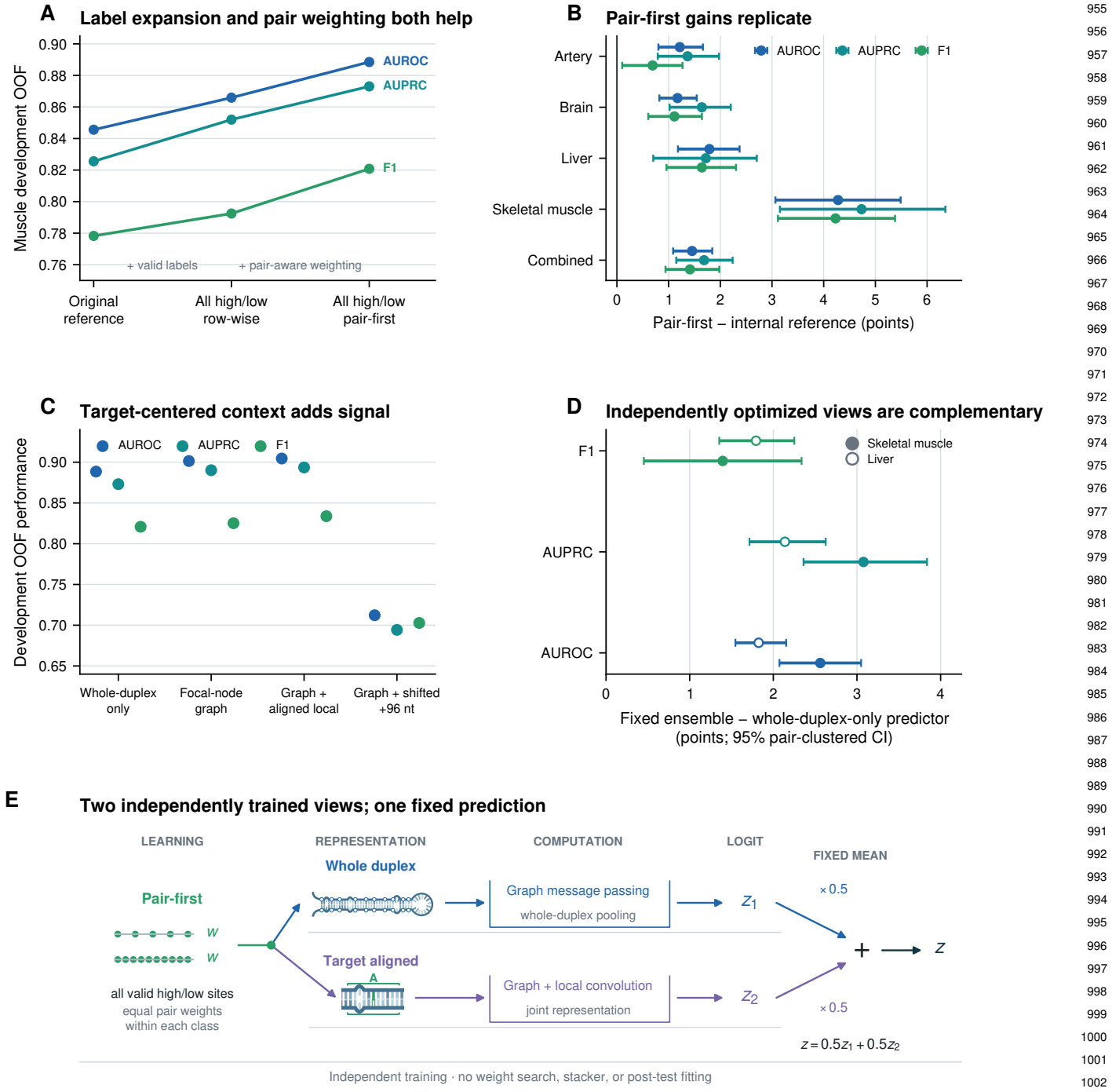


Fig. 5. Pair-first learning and target-centered representation provide complementary gains. (A) Same-fold ablations separate recovery of additional valid high/low labels from pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Comparisons test whole-duplex pooling, explicit retention of the target representation, and addition of an aligned local window. The control shifts that window by 96 nt in an otherwise identical model. (D) A fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. Effects in B and D are percentage points. (E) Independently trained whole-duplex-only and graph-plus-local models produce log-odds scores, combined without weight search or post-test fitting. RNA illustrations are conceptual. All panels use development refits; their absolute scores are not the supplied-checkpoint test scores.

summary outcomes and predicted structures as surrogates for in vivo conformations. The canonical labels are balanced and exclude intermediate observations; natural-prevalence analyses remain secondary. eCLIP is cell-line-specific and incompletely maps repeat-associated binding. Nonhuman transfer mixes differences in lineage, substrate class, assay, and labeling.

Component attribution comes from adaptively selected development comparisons without multiplicity adjustment; early pooled-score residual screens additionally require fully nested replication. Tissue comparisons share substrates and are not independent cohorts. The frozen test assesses unseen pairs within the benchmark construction, not transportability to an independently generated human cohort (16).

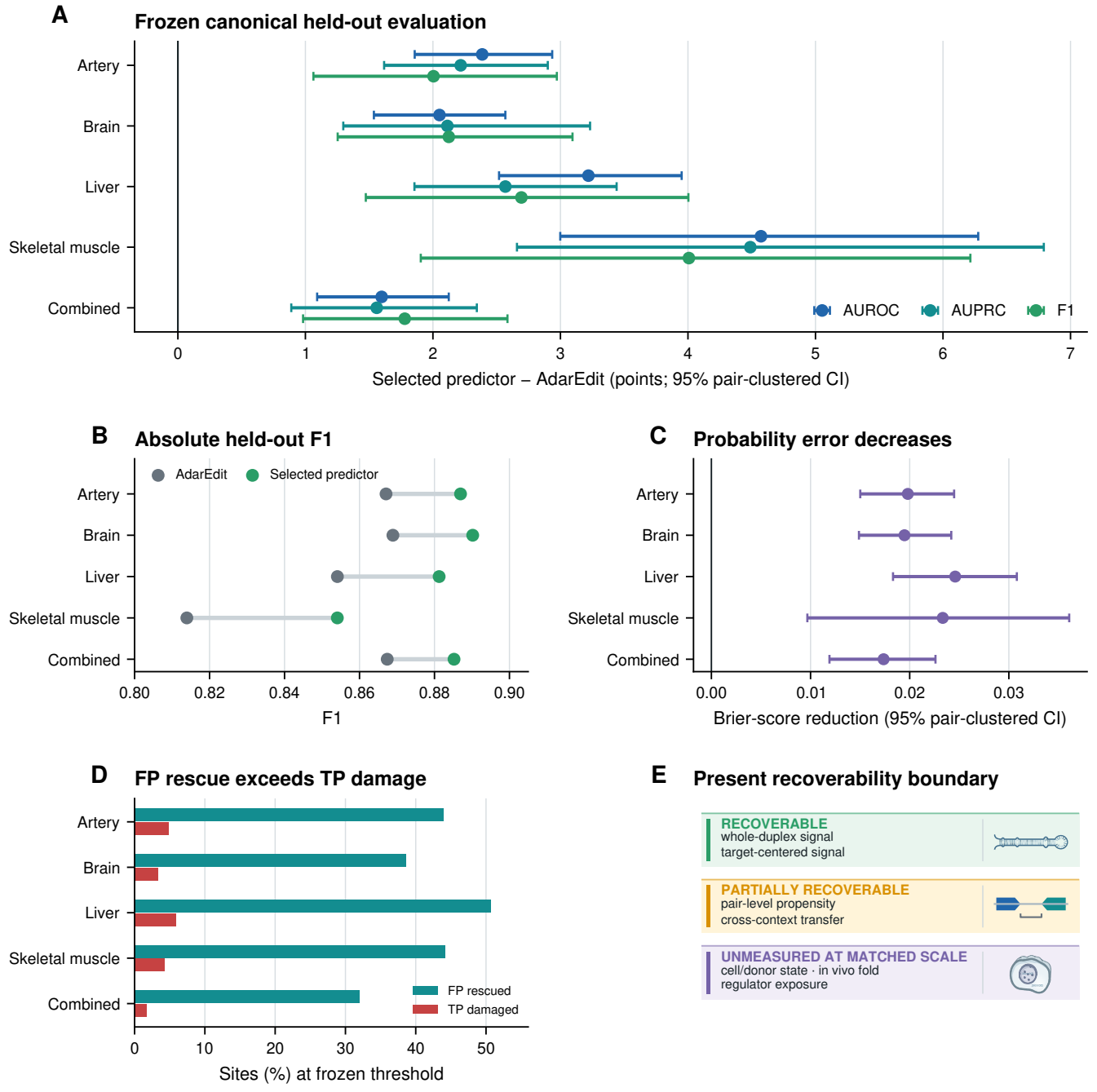


Fig. 6. The selected combination improves prediction on unseen benchmark pairs. (A) Selected predictor minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; effects are percentage points, with 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score; positive values indicate lower probability error. (D) Baseline false positives corrected and baseline true positives converted to false negatives at frozen validation-selected thresholds, each as a percentage of its own baseline group. (E) Synthesis of recoverable, partially resolved, and unmeasured information. The unmeasured tier identifies candidate measurements, not the causes of all residual error; icons are conceptual. Combined is pooled. Quantitative panels use the frozen canonical held-out evaluation.

The next biological test is to measure editing and candidate regulators on the same RNA substrates in the same donors or cells. Candidate measurements include transcript processing, ADAR isoform and interferon state, localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation. They are not a proven sufficient set. Their value should be judged by whether they improve prediction of editing differences at the same sites across biological contexts, beyond what sequence, predicted pairing, and genomic organization already predict.

Materials and Methods

Search design and evidence classification. The adaptive search grouped candidates by intrinsic substrate properties, transcript/cellular context, and target-local selection, with separate axes for dependent observations and model sufficiency. The latter covered richer supervision, cross-context sharing, richer/pretrained representations, and capacity or integration. Priorities reflected biological rationale, information beyond the

reference, measurement compatibility, leakage risk, and effective sample size. Development branches used pair-grouped comparisons against a compatible reference, with randomized, shifted-location, or matched-capacity controls where relevant.

Selected branches met recorded selection criteria and entered the final predictor; supported findings were retained without selection; limited effects were metric- or context-specific. Not supported denotes failure of the tested formulation. Deferred branches lacked suitable data or a validated resource; superseded branches were not pursued after stronger evidence emerged. These adaptive comparisons are not multiplicity-adjusted confirmation, and the full program was not preregistered. SI Appendix, Tables S2 and S10 distinguish controlled comparisons, exploratory screens, descriptive characterization, and untested branches. Only the frozen combination entered the final canonical evaluation.

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human sites were labeled edited at EditingIndex $\geq 15\%$, nonedited at EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. We verified pair identities and the fixed 566/139/179 train/validation/test assignment before analysis. Combined was used only as its published pooled endpoint.

Development protocol. Discovery used the original train and validation pairs in 5 outer folds by pair ID. Neural comparisons excluded the outer fold from training, preprocessing, checkpoint choice, and threshold choice; one other fold provided internal validation. Early residual screens instead fitted pair-disjoint second-stage models to pooled first-stage OOF logits. Their meta-training scores could depend on the current outer fold through first-stage fitting, so they are exploratory, not fully nested estimates (SI Appendix, section A). Later outer-isolated gates and foundation fusion used fold-specific first-stage fits. Supplied checkpoints were reproduced using aggregate test metrics before discovery; candidate development did not access the canonical test.

Models. The first component retained the compact AdarEdit graph attention encoder. Each epoch allocated equal high/low class counts, sampled pairs uniformly among those eligible for each class, and then sampled a site within each selected pair. The canonical-row update budget was unchanged. The second component was trained independently with the same graph encoder. It added a 65-position target/helix-partner-aligned representation encoded by two convolutional layers, then fused local and graph embeddings. The selected probability was $\sigma[(\operatorname{logit} p_{\mathrm{whole}} + \operatorname{logit} p_{\mathrm{augmented}})/2]$, where the two p values are component probabilities and σ is the logistic function. Weights were fixed at 0.5/0.5 with no weight search. The shifted-window control preserved architecture and optimization while moving the local center 96 nt. SI Appendix gives the feature definitions and remaining protocols.

Statistics and final evaluation. Metrics were AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, MCC, and Brier score. Thresholds maximized validation F1 within each run. Pair-clustered bootstraps resampled Alu pairs with all their sites: development comparisons used 1,000 or 2,000 replicates, the post-hoc within-duplex analysis used 5,000, and the final comparison used 2,000. These intervals condition on the fitted models and do not include variability from rerunning training or adaptive selection. Code, inputs, weights, epochs, thresholds, and checkpoint hashes were frozen before the selected predictor's held-out evaluation; no subsequent model selection or tuning occurred. Natural-prevalence and matched-recall analyses are secondary (SI Appendix).

AI-assisted preparation. OpenAI Codex assisted with manuscript editing, source-code review, and figure-rendering code. RNA and cell illustrations were adapted from credited DBCLS/TogoTV and Servier Medical Art SVGs (SI Appendix, Artwork credits). Earlier GPT Image 1.5 illustrations are excluded. Author confirmation of the complete tool/model-version disclosure and of journal eligibility for AI-assisted diagram code is pending.

Data, Materials, and Software Availability. Code, experiment configurations, registries, source tables, figure scripts, retained derived artifacts, frozen checkpoints, and provenance are available at https://github.com/Scientific-Computing-Lab/ADARPro. Third-party downloads and deterministic matrices omitted from GitHub are listed with source URLs, roles, checksums, and regeneration instructions in the data manifest. The AdarEdit benchmark is available from https://github.com/Scientific-Computing-Lab/AdarEdit. No new human participants were recruited and no new primary sequencing data were generated.

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